HARVEY BENEDICT PARENTE

harveyparente@gmail.com | linkedin | Portfolio

EDUCATION

FAR EASTERN UNIVERSITY - INSTITUTE OF TECHNOLOGY *Padre Paredes St, Sampaloc, Manila*
Bachelor of Science ***Expected September 2027***
Major in Computer Science, specialized in Data Science
Cumulative GPA: 3.xx

Relevant Coursework*:* Object Oriented Programming; Data Structures and Algorithms; Database Systems; Modeling and Simulation; Data Visualization; Statistical Analysis and Modeling; Advanced Machine Learning; Large Language Models; Cybersecurity

WORK EXPERIENCE

Institute for Data and Statistical Analysis *Sta. Mesa, Manila, Philippines*
Data Science Intern *February 2023 – May 2023*

- Analyzed 82,000+ records using IBM SPSS and statistical modeling techniques to identify trends that informed institutional decision-making
- Cleaned and validated datasets in Microsoft Excel to ensure the accuracy and reliability of analysis results
- Collaborated with cross-functional teams by performing analytical work on departmental data, delivering insights that supported improved institutional outcomes
- Presented modeling results to department heads and directors through clear reports and presentations to support decision-making

PROJECTS

GeneScope: Predicting Genetic Testing Utilization in the Philippines Using Binary Logistic Regression
Python, scikit-learn, Flask, TanStack Start (React/TypeScript), PostgreSQL | Netlify/Cloudflare, Render, Neon
October 2025 - October 2026

- Developed a predictive system with a four-member team in partnership with an industry healthcare company to classify genetic test type (targeted vs. comprehensive) from clinical, geographic, and institutional factors, supporting equitable healthcare planning in the Philippines
- Engineered features from 447 de-identified patient records and tuned three models using GridSearchCV and 5-fold cross-validation, correcting a 3:1 class imbalance through class weighting
- Achieved 82.22% test accuracy and a 0.8275 cross-validated ROC-AUC with an interpretable Logistic Regression model
- Identified disease category as the strongest predictor (Pediatrics OR = 22.43) and quantified a Mindanao neurology access gap (OR = 0.46) to guide regional resource allocation

EXTRACURRICULAR ACTIVITIES

National Service Reserve Corps (NSRC) *Padre Paredes St, Sampaloc, Manila*
Chapter Leader *November 2024 – April 2025*

- Led an elected chapter leadership role, coordinating members and activities for disaster and humanitarian relief programs
- Partnered with the Philippine Coast Guard to organize water survival training that prepared participants for emergency response
- Directed community outreach in Agoncillo, Batangas, delivering aid to families affected by the Taal eruption
- Mobilized volunteers and resources to support relief operations

ADDITIONAL

Technical Skills: Advanced in Python, Proficient in SQL
Programming Languages: Java, C++, MATLAB, PHP, HTML/CSS, XML, Kotlin
Certifications: Python, PMI Project Management Ready, Android Development with Kotlin, Linux Essentials
Languages: Filipino (native); English (fluent); Mandarin (beginner)